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Roll No. 1972075..... 34 Page

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**MCA (First Semester) EXAMINATION,
Dec. 2022**

Paper – MCA-103

**PROGRAMMING AND PROBLEM
SOLVING USING 'C' AND C++**

Time : Three Hours

Maximum Marks : 60

Minimum Pass Marks : 21

Note—Attempt *all* questions. Question no. 1 is compulsory. Question 2 to 6 carry equal marks.

1. Attempt any *five* questions in brief—
 - (i) What is Robustness ?
 - (ii) What do you understand by top down design ?
 - (iii) What do you mean by ternary operator ?

P.T.O.

(2)

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- (iv) What do you understand by C++ ?
- (v) What is function in C ?
- (vi) What do you mean by classes ?
- (vii) What is pointer ?
- (viii) What is the input of C ?
- (ix) Write any two characteristics of good program.
- (x) Define naming variables.

Unit-I

- 2. (a) What is analysis and design ?
- (b) Write an algorithm to calculate and print table of 2.

Or

- (a) What is flow chart ? Write about any program with the help of flowchart.
- (b) How to identify the problem ? Write about the Technique ?

Unit-II

- 3. (a) Define the following terms—
 - (a) Data types.
 - (b) Constants.
 - (c) Variable.
 - (d) Operators.

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- (b) What is ternary operator ?

Or

- (a) Define the following control constructs—
 - (a) if-else.
 - (b) While
 - (c) do-while.
- (b) What is Arrays ? Write any programme with use an Array ?

Unit-III

- 4. (a) What is call by value and call by reference ?
- (b) What is return statements ?

Or

- (a) Define tail recursion.
- (b) What is structures and union ?

Unit-IV

- 5. (a) What is Basic concept of object oriented programming ?
- (b) Explain the term of Inheritance.

Or

- (a) Explain function-inline functions ?
- (b) What is friend function ?

P.T.O.

Unit-V

6. (a) What is operator overloading ?
(b) Define virtual base class.

Or

- (a) Define pure virtual function.
(b) Explain about the Dynamic memory allocation operators.